

Language Modeling

- *Language model*: a probabilistic mechanism for generating text
 - Or equivalently, a probabilistic model that defines a distribution of text
- Claude Shannon (“Prediction and entropy of printed English”, 1951) first investigated the problem of the extent to which the next word is predictable

Language Modeling

- Shannon's Game: predict the next letter given the former ones:

(1) THE ROOM WAS NOT VERY LIGHT A SMALL OBLONG

(2) ----ROO-----NOT-V-----I-----SM----OBL-----

(1) READING LAMP ON THE DESK SHED GLOW ON

(2) REA-----O-----D-----SHED-GLO--O--

(1) POLISHED WOOD BUT LESS ON THE SHABBY RED CARPET

(2) P-L-S-----O---BU--L-S--O-----SH-----RE--C-----

Language Modeling

- Shannon's game allows estimating the unpredictability, or the information in each letter and word
- Can computers do the same?
 - This could be an interesting benchmark for language technology
- If so, what is language modeling useful for?
- Is there psychological evidence that humans are sensitive to the predictability of the next word?
 - There is

The Language Modeling Problem

- **Goal:** given a sequence of words, what's the distribution of the next word given the “prefix”

$$P(x_n | x_1, \dots, x_{n-1})$$

- More formally:
 - Setup: assume a finite set of words (vocabulary) V
 - Data: a training set of examples (finite strings over V)
 - Problem: estimate the probability distribution over all finite strings over V

$$\sum_{x \in V^+} P(x) = 1$$

The Language Modeling Problem

- From the joint to the conditional distribution:

$$P(x_n | x_1, \dots, x_{n-1}) = \frac{P(x_1, \dots, x_n)}{P(x_1, \dots, x_{n-1})}$$

- From the conditional to the joint distribution:

$$P(x) = P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | x_1, \dots, x_n)$$

Language Modeling

- But what is this distribution?
 - Is there a “ground truth” distribution?
 - If not, what are useful distributions? what should we aim to capture?
- It is not clear whether there is a sense in which there is a “ground truth” distribution. Surely, it needs to depend on the domain, context and speaker/writer

Language Modeling

- At a minimum, probabilities should broadly indicate plausibility of sentences:
 - $P(I \text{ saw a van}) \gg P(\text{eyes awe of an})$
 - Not only grammaticality: $P(\text{artichokes intimidate zippers}) \approx 0$
 - In principle, plausibility depends on the domain, context, speaker
- The minimal requirements reflect the classic approach to LMs

Language Modeling

- Using LMs to generate text makes much stronger assumptions, and essentially aims to reduce all dimensions of language into a single number
- This is its strength - we will see it greatly simplifies the task of NLP
- But also its weakness - language is fundamentally multi-dimensional in its core and much is lost by placing different strings on a scalar scale

Learning Language Models

- Goal: Assign useful probabilities $P(x)$ to sentences x
- Input: many observations of training sentences x
- Output: system capable of computing $P(x)$
- The straightforward approach is viewing it as density estimation over the space of finite strings over a given alphabet

Trivial Language Model

- Simplest option: empirical distribution over training sentences...

$$p(x_1 \dots x_n) = \frac{c(x_1 \dots x_n)}{N} \text{ for sentence } x = x_1 \dots x_n$$

- Problem: does not generalize (at all)
- We need to assign non-zero probability to previously unseen sentences!